Ethan Leiva Pardo

ROBOTICS & AI ENGINEER | IoT | AUTOMATION | R&D;

ethant.leiva@gmail.com · +56 9 3076 4786 · linkedin.com/in/ethanleiva · ethan.tecbit.cl
Santiago, Chile

PROFESSIONAL PROFILE

Computer Engineer with 5+ years of experience building and integrating technology across robotics, artificial intelligence, IoT, automation and industrial hardware. Experience leading multidisciplinary technology projects from architecture and prototyping through implementation, technical support and knowledge transfer. Strong hands-on background in **ROS 2, Python, Computer Vision, edge computing, ESP32, NVIDIA Jetson, web systems and IoT**, with a focus on turning real operational problems into deployable technology solutions.

CORE TECHNICAL SKILLS

Robotics & Edge AI	ROS 2 · NVIDIA Jetson Orin · ESP32 · Arduino · Duckietown · Kondo
AI & Computer Vision	Artificial Intelligence · Computer Vision · OpenCV · Local LLMs · Image processing · Facial recognition
Software Engineering	Python · Java · C/C++ · Flask · Node.js · React · Ionic · APIs · Web systems
IoT & Automation	Node-RED · Telemetry · Microcontrollers · Remote monitoring · Networked control · Industrial diagnostics
Leadership	Technical project leadership · Solution architecture · Client interaction · Training · Technology transfer

PROFESSIONAL EXPERIENCE

Technical Specialist — Industrial Board Repair & Diagnostics

HOD Vending | Logistics / Distribution | Jul 2026 – Present

- Established diagnostic and repair processes for industrial electronic boards, becoming the first technician dedicated to this function within the company.
- Diagnose and recover vending-machine boards, creating technical procedures, documentation and repair criteria to standardize work.
- Train field technicians and transfer troubleshooting knowledge to strengthen diagnostic and failure-resolution capabilities.
- Develop internal tools for automation, inventory control and telemetry prototypes to improve technical and operational processes.

Co-founder & Technology Project Director

Tecbit Limitada | Technology | Aug 2024 – Present

- Design and deliver customized technology solutions integrating frontend, backend, databases, APIs and cloud services.
- Lead automation and digitalization initiatives aimed at reducing manual work, operational time and costs; selected implementations achieved **up to 60% efficiency improvements** and **up to 25% reductions in operating costs**.
- Define solution architectures, technology choices, implementation priorities and production environments.
- Integrate systems and services to centralize information and improve operational decision-making; manage projects end-to-end and work directly with clients.

Technology Project & Technology Transfer Lead

Duoc UC — Center for Innovation and Technology Transfer (CITT) | Aug 2023 – Jan 2026

- Led multidisciplinary technology projects combining software, hardware, robotics, AI, IoT, automation and computer vision.
- Principal creator of **Bitty**, a social-impact robotic platform integrating AI, computer vision, ROS 2, software and human-robot interaction.
- Developed Bitty V1/V2 and participated in autonomous robotics projects using Duckietown/Duckiebot and Kondo platforms.
- Delivered institutional technology solutions including CITTPass, Cibervoluntarios, AllTV Duoc and All Air Controller.
- Trained and mentored students in programming, robotics, electronics and technology prototyping; organized workshops and technology-transfer activities.

Software Developer

Austranet | Dec 2021 – Mar 2022

- Developed AI-based solutions for summarizing and simplifying Chilean legislation and supporting document-analysis workflows.
- Implemented an AI solution for automated complaint classification to improve operational efficiency.

Software & IoT Developer

Ecolav | Sep 2020 – Feb 2021

- Developed core software for detergent vending machines, integrating control and operational functionality.
- Implemented IoT capabilities for remote monitoring and supervision of dispenser operation.

SELECTED R&D; PROJECTS

Bitty 2.0 — Robotics, AI & Computer Vision

Robotics · ROS 2 · NVIDIA Jetson Orin · ESP32 · Python · OpenCV · Flask · Node.js · React · Ionic · Local LLMs

Multidisciplinary robotic platform for experimentation, learning and technology transfer. Uses ROS 2 for coordination, Jetson Orin for intensive AI/vision processing and an ESP32-based local server; integrates computer vision, local language models, web/mobile control and human-robot interaction, including Meta Quest 3 experimentation.

ALL TV DUOC — IoT Automation

IoT · Automation · Remote Control · Energy Efficiency

Corporate project executed through Tecbit to automate remote power control of institutional displays across Duoc UC sites, reducing operating hours and improving energy efficiency.

Bitty Chat — AI Assistant

AI · Conversational Interface · Institutional Information

Conversational AI interface providing institutional information, transportation status, weather and automated responses through AI models.

Autonomous Robotics & Technology Transfer

Duckietown · Duckiebot · Kondo · Robotics Training

Hands-on work with autonomous-driving robotics platforms and humanoid robots, combined with training and technology transfer for students across multiple campuses.

IoT Community Experiences

Sensors · AI · Microcontrollers · Networked Control · Automation

Designed and deployed interactive IoT/AI systems, including a sensor-driven Halloween experience with coordinated lights, sound and mechanical effects, and an annual Christmas light show synchronized with music and remotely controlled through mobile devices.

EDUCATION

Engineering in Computer Science / Informatics — Graduated	Duoc UC, Antonio Varas Campus | 2023–2025
Civil Engineering in Computer Science — Incomplete	Universidad Técnica Federico Santa María | 2021–2022
Technical Secondary Education	Liceo Politécnico Andes | Graduated 2020

ADDITIONAL EXPERIENCE & RECOGNITION

- **Programming Instructor — Humana Consultores:** project-based programming workshops for children using Scratch.
- **Leadership & mentoring:** technical training, student mentoring, workshops, competitions and technology-transfer activities.
- **Languages:** Spanish (native) · English (intermediate).
- **Mobility:** Class B driver's license and own vehicle.
- **External recognition:** Bitty featured in coverage associated with Congreso Futuro 2025 / BioBioChile and the Arena III robotics event.

Portfolio: ethan.tecbit.cl · **LinkedIn:** linkedin.com/in/ethanleiva